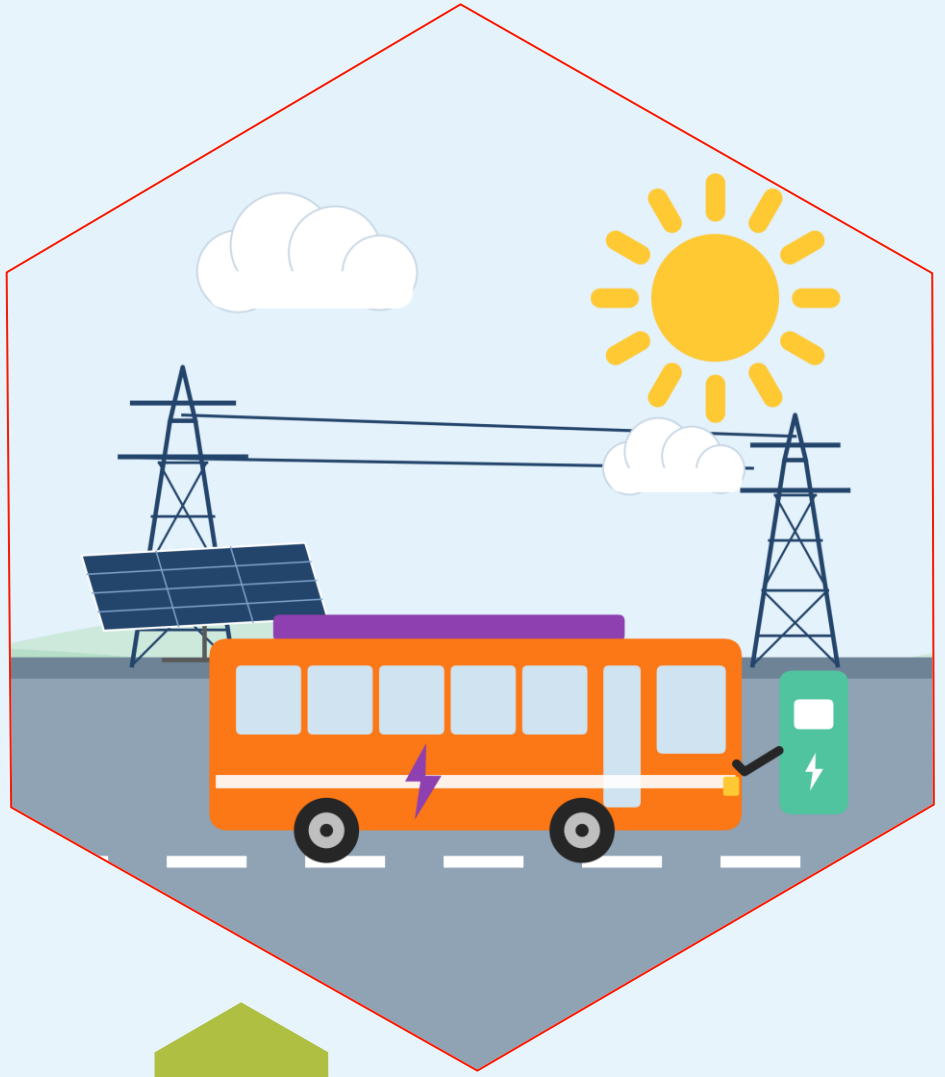
Lecture 5:
Problem Framing & Modelling II



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Disclaimer: this lecture references material adapted from Kosky, Balmer, Keat & Wise, Exploring Engineering (5th ed.), and from the instructor's own course slides, with modifications. No authorship of third-party material is claimed.

Last Week, in One Breath

Evidence — we sort what we hear into fact, inference, speculation, and opinion.

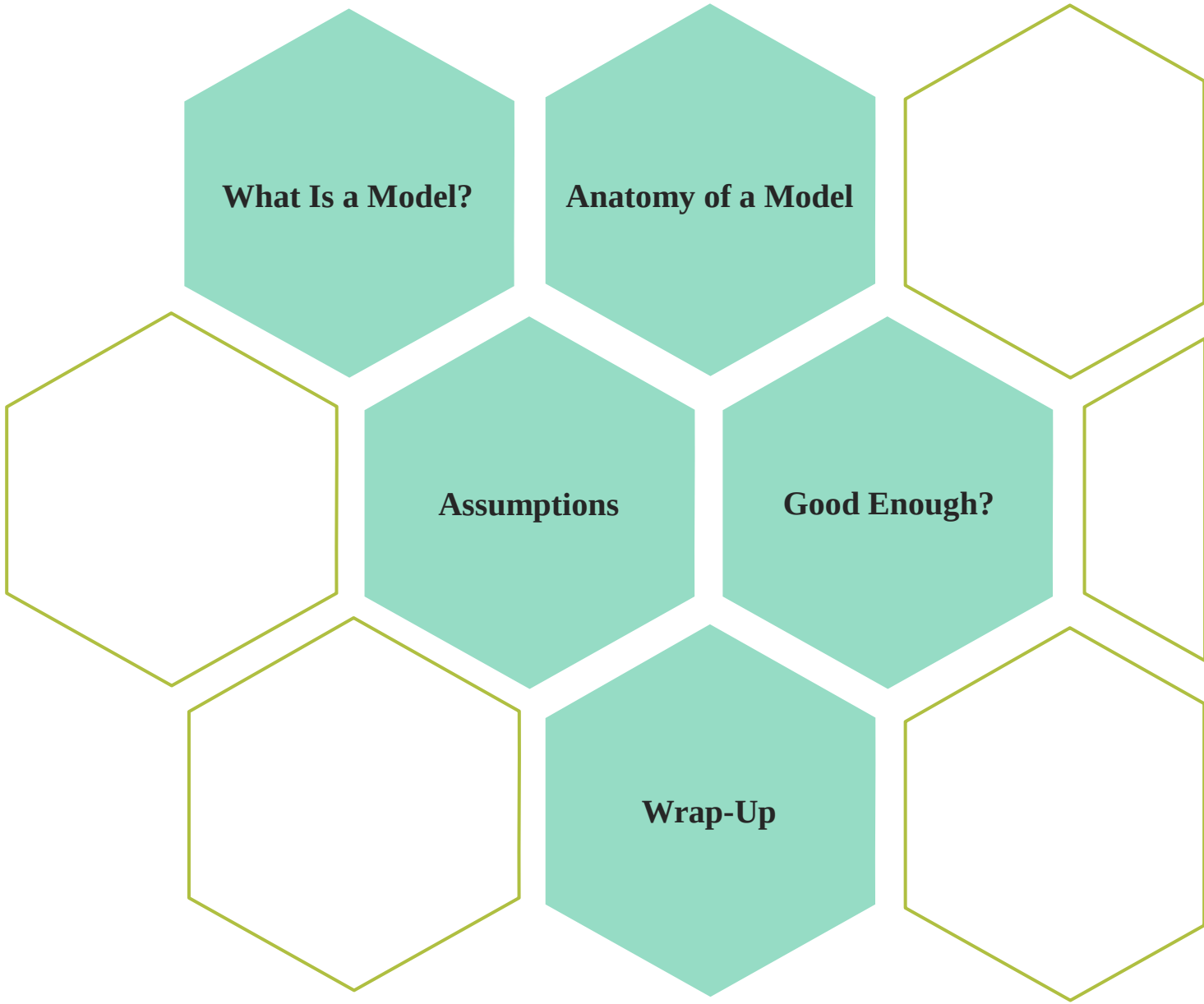
Needs, not solutions — a good problem statement names no device, app, or vehicle.

Variables — measurable quantities with units: performance, design, and context.

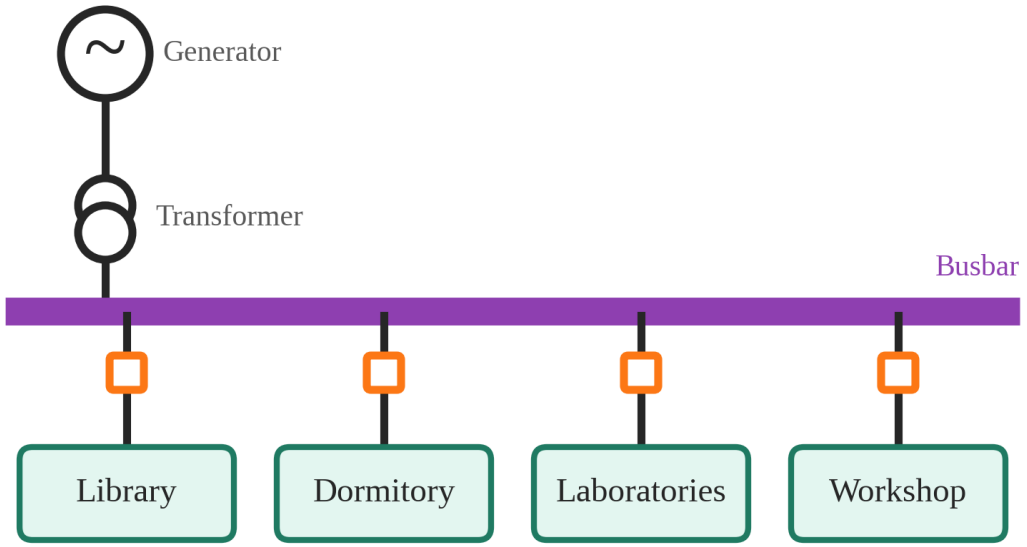
Performance — stakeholders define “working well”; we write it down in numbers.

Today — turn a framed problem into a simple model, and say what it assumes.

Agenda

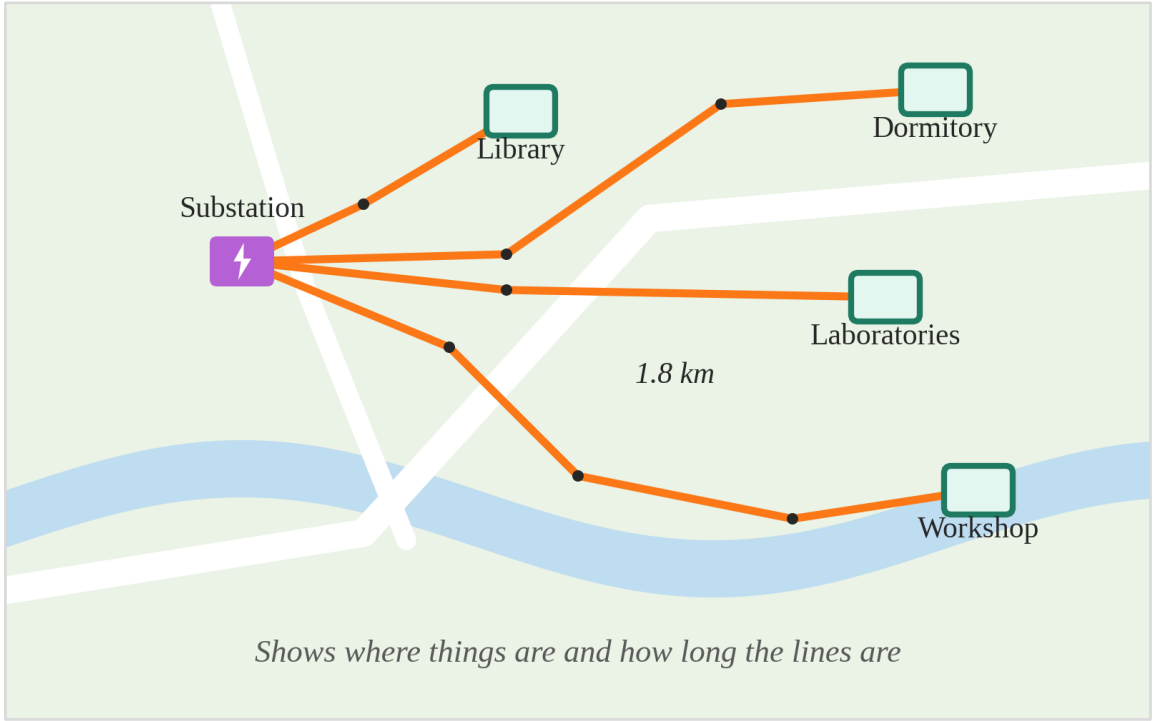

What Is a Model?
Anatomy of a Model
Assumptions
Good Enough?
Wrap-Up

One Grid, Two Maps



Shows how everything is connected, not where or how far

A · Single-line diagram



Shows where things are and how long the lines are

B · Geographic layout

Same campus, same grid. Show of hands: ***A or B, which map is the right one?***

Which Map Is Right? It Depends

MAP A · SINGLE-LINE DIAGRAM

Answers connection questions

If the library feeder trips, which buildings lose power?

Where should we place a switch?

Which breaker isolates a fault?

Leaves out: distances, cable sizes, time.

MAP B · GEOGRAPHIC LAYOUT

Answers location questions

How long is the cable to the workshop?

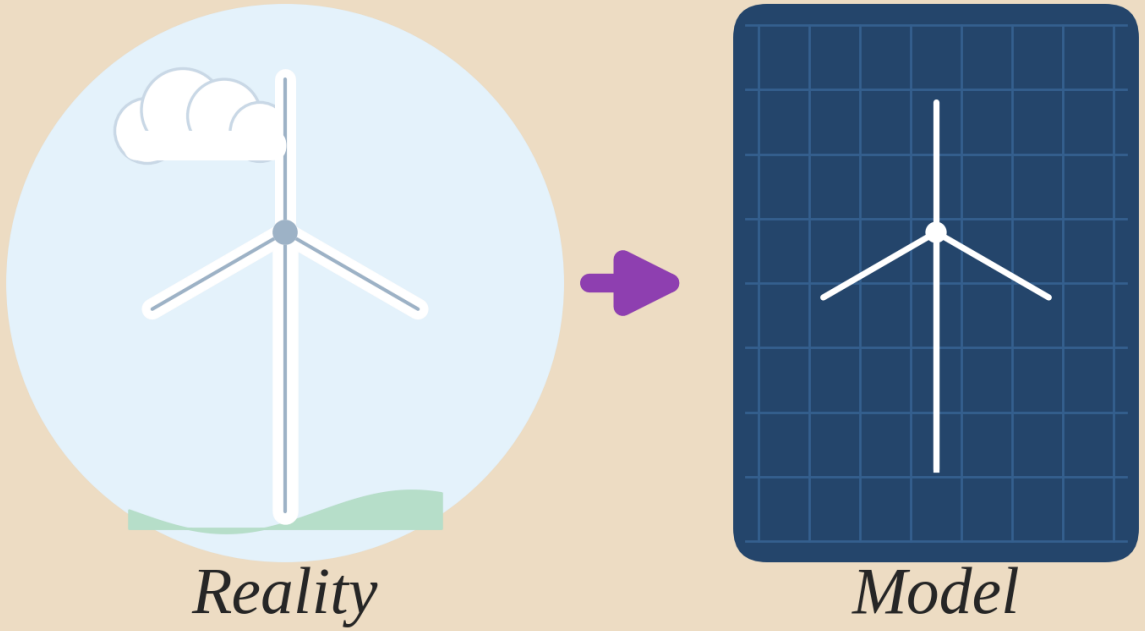
Which lines cross the river or the road?

Which building is closest to the substation?

Leaves out: switching, protection, capacities.

Neither map is the grid.
Each is a model: a simplified picture made to answer one kind of question.

PART 1



Reality *Model*

What Is a Model?

A simplified stand-in for reality, built to answer a question.

A Model Is a Purposeful Simplification

A **model** is a simplified representation of a real system, built to answer a specific question.

KEEPS

What matters

For the bus range question: battery energy, distance, energy used per km.

DROPS

What barely matters

Paint colour, number of seats, the exact shape of every turn.

PURPOSE

The question

“Can one charge cover a day on the route?”

“All models are wrong, but some are useful.” (George Box, statistician)

Four Types of Models

PHYSICAL

A real copy

A smaller or simpler real object.

A wind-turbine blade tested in a wind tunnel.

GRAPHICAL

A diagram

A drawing of parts and links.

A single-line diagram, or an energy-flow diagram of a car.

MATHEMATICAL

Equations

Relations between variables.

Ohm's law and the power law for an electric circuit.

COMPUTATIONAL

A calculation

Rules stepped through on a computer.

A spreadsheet of hourly electricity demand.

Real projects mix several types: a diagram to organise, equations to calculate, a spreadsheet to run the numbers.

Examples are ours. Energy-flow model of a car: Kosky et al., Ch. 4. Circuit model: Kosky et al., Ch. 10.

One System, Three Models

System: the campus electricity supply (illustrative)

Model	Type	It answers	It leaves out
Single-line diagram	Graphical	Which buildings lose power if one feeder trips?	Distances, power levels, time
Energy balance: supply = demand + losses	Mathematical	Does daily supply cover daily demand?	When in the day the demand occurs
Hourly demand spreadsheet	Computational	At which hour does the peak occur?	Network layout, voltages

Same system, three models. The question decides which one to use.

Match the Question to the Model

In pairs, pick the model type that fits each question best: physical, graphical, mathematical, or computational.

A Will the electric bus finish its route on one charge?

B Which buildings lose power if one feeder trips?

C How does a dormitory’s temperature change over a winter day?

D Will a new turbine blade survive strong wind?

E Where do riders queue at the busiest bus stop?

F How many kWh does the faculty building use in a day?

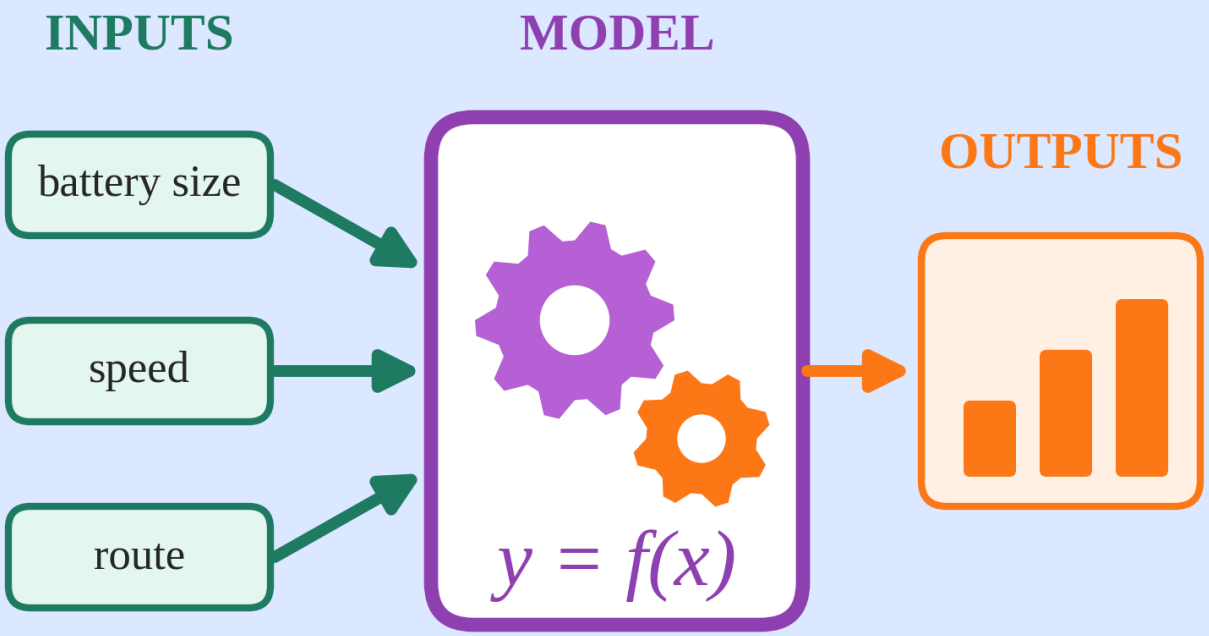
More than one type may fit. Be ready to say why.

Match the Questions: Answer Key

	Best fit	Why
A	Mathematical	Energy used = energy per km × distance, compared with the battery energy.
B	Graphical	A single-line diagram shows which feeder supplies which building.
C	Computational	Temperature changes hour by hour, so we step through time.
D	Physical	A scaled blade in a wind tunnel tests real airflow and loads.
E	Graphical	A sketch of the stop shows where people stand and move.
F	Mathematical	Power (kW) × time (h) = energy (kWh), summed over the loads.

Several types can fit. What matters is that you can say why.

PART 2



Anatomy of a Model

Inputs, parameters, relationships, and outputs.

Four Parts of Every Model

INPUTS

What we feed in

Values we choose or can change.

Battery size, speed, route length.

PARAMETERS

What we hold fixed

Numbers that describe the system and its setting.

Energy use per km, usable share of battery.

RELATIONSHIPS

The rules

Equations or steps that link the numbers.

Energy = energy per km × distance.

OUTPUTS

What it predicts

The results we judge.

Range (km), energy per day (kWh).

A model is a machine: numbers go in, a rule is applied, numbers come out.
Every number needs its units.

Last Week’s Variables Become Model Roles

Lecture 4 variable kind	Role in the model	Electric campus bus (illustrative)
Performance	Output	Range per charge (km), energy per day (kWh/day)
Design	Input	Battery capacity (kWh), number of buses
Context	Parameter	Route length (km), outside temperature (°C), hours at the depot charger

Last week’s variable list is already the skeleton of this week’s model.

Need–Know–How–Solve, Applied to the Bus

NEED

The variable we want

Range of one charge (km).

KNOW

What we already know

Battery capacity, usable share, energy per km, route length.

HOW

The model and assumptions

Range = usable energy ÷ energy per km.

SOLVE

Do the calculation

Compute, check units, interpret.

Kosky et al. note that the How also includes the assumptions you make to solve the problem. Last week gave us the Need; this week we build the How.

Today's focus: the How, and the assumptions hidden inside it.

Method: Kosky et al., Ch. 2 (The Need-Know-How-Solve Method). Example values are illustrative.

A First Model: the Electric Campus Bus

THE MODEL

Energy for a trip

$$E = e \times d$$

Range on one charge

$$R = (u \times C) \div e$$

Symbol	Meaning	Unit
C	Battery capacity	kWh
u	Usable share of the battery	(no unit)
e	Energy use per km	kWh/km
d	Distance driven	km
R	Range on one charge	km
E	Energy for a trip	kWh

Units check, Kosky style: (u × C) [kWh] ÷ e [kWh/km] = R [km]

Putting Numbers In

160 kWh

Usable energy

$u \times C = 0.8 \times 200\ kWh$

133 km

Range on one charge

$R = 160\ kWh \div 1.2\ kWh/km$

96 km

Daily need

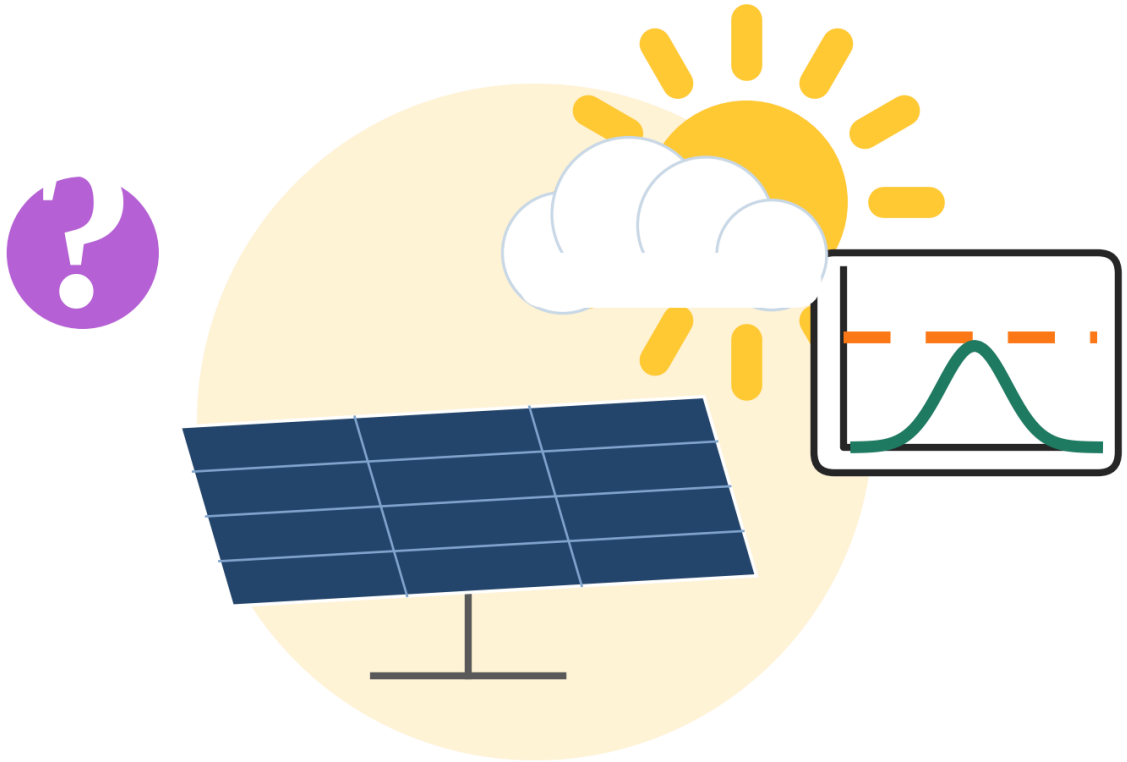
$8\ loops \times 12\ km$

The model says: one charge covers the day, with a margin of about 37 km.

But how far should we trust that number? It depends on the assumptions.

Illustrative values, not data.

PART 3



Assumptions

What we ignore, what we fix, what we guess.

Three Kinds of Assumptions

SIMPLIFYING

What we ignore

Hills, traffic lights, stops.

Differences between drivers.

DATA

What we fix

e = 1.2 kWh/km.

C = 200 kWh, u = 0.8.

BOUNDARY

What we leave outside

Passengers, charger, and the grid.

Only the bus is in the model.

Lecture 2 idea, again: the boundary is a choice, and so is every assumption.

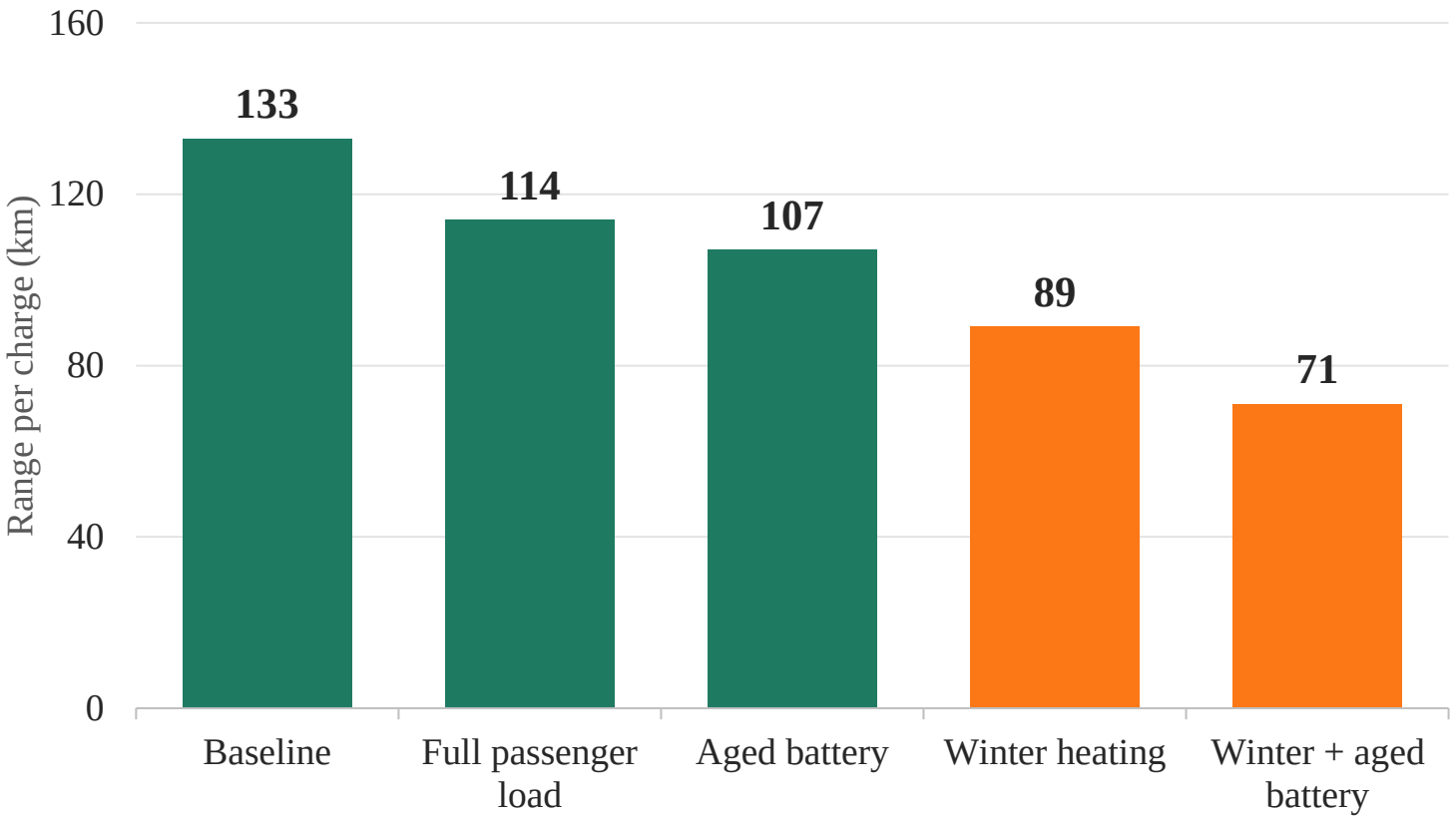
Idea from Kosky et al., Ch. 4 (“beware of your assumptions”) and Ch. 2 (the How includes the assumptions).

Hidden Assumptions in the Bus Model

Model choice	Hidden assumption	If it is wrong
e = 1.2 kWh/km	The same in every season	Cabin heating raises e, so range falls
e = 1.2 kWh/km	Flat route, similar passenger load	Hills and full buses use more energy
u × C = 160 kWh	The battery never loses capacity	An ageing battery stores less
Start with a full battery	Depot charging always works	A missed charge cuts the day short

None of these is written in the equation. That is why we write them down.

Same Model, Different Assumptions



DAILY NEED

96 km

Covers the day: baseline, full load, aged battery.

Falls short: winter heating, and winter with an aged battery.

Cases: e = 1.4, e = 1.8 kWh/km, usable energy 128 kWh.

"One charge covers the day" holds only while the assumptions hold.

Illustrative values, not measured data.

A 5 kW Array Is Not 5 kW All Day

HIDDEN ASSUMPTION

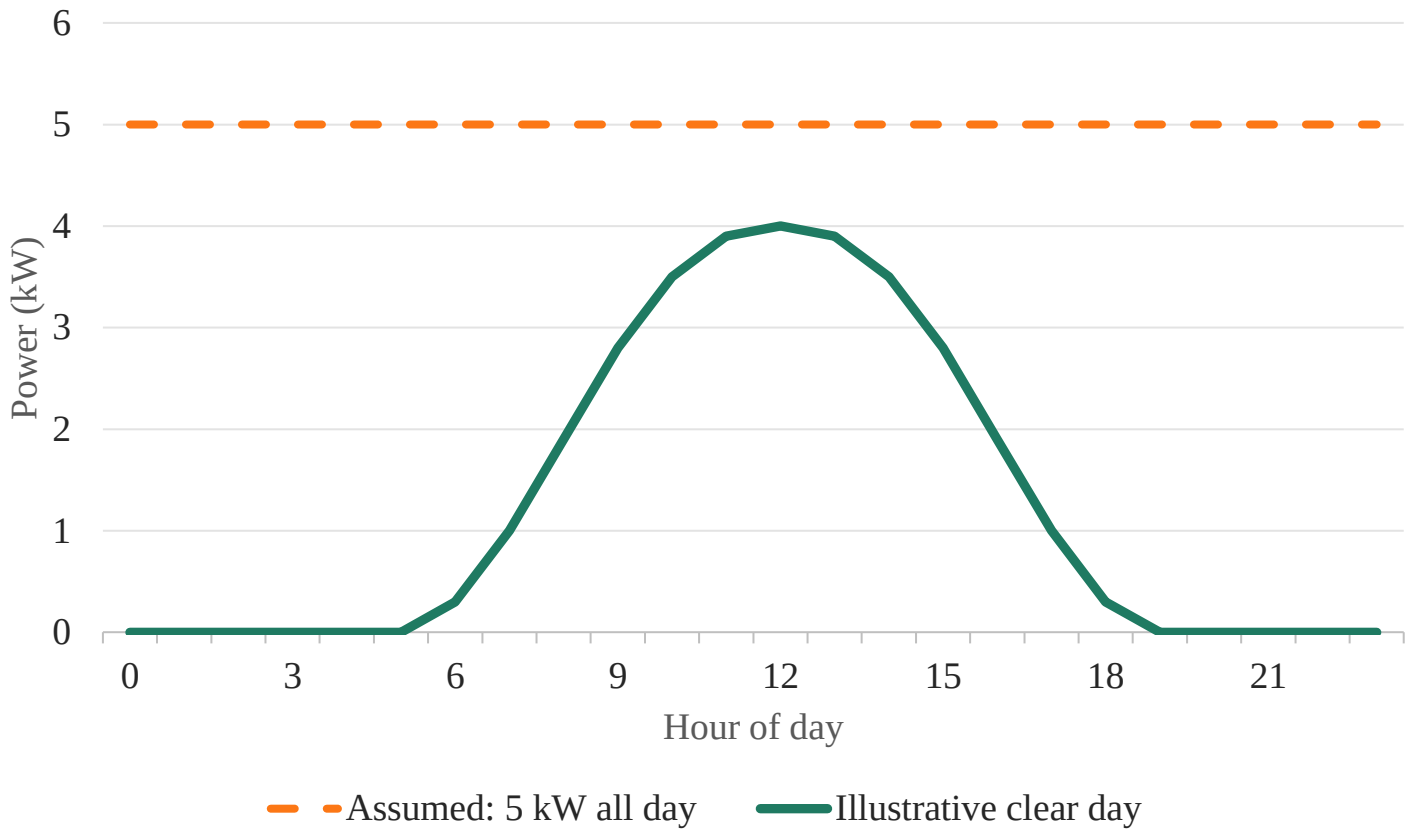
The sun shines at full strength all day

Naive model: 5 kW × 24 h = 120 kWh per day.

Clear-day profile: about 31 kWh per day.

The naive model is almost four times too high.

Note the units: kW is power, kWh is energy.



Before trusting a number, ask which conditions it silently assumes.

Illustrative profile, not measured data.

Writing Assumptions Down

COURSE TEMPLATE

We assume **[assumption]** because **[reason]**. If it is wrong, **[effect on the result]**.

Example (bus)

We assume 1.2 kWh/km in all seasons because we only have autumn test data. If it is wrong, winter range could fall below the 96 km daily need.

Quick check

Could a teammate tell what you ignored, fixed, or left outside? Could you test it?

An assumption that is written down can be checked. A hidden one cannot.

Template is a course adaptation. Audit-trail idea: Kosky et al., Ch. 2.

Spot the Hidden Assumptions

In pairs: find at least one hidden assumption in each statement, then rewrite it with the template: we assume …, because …; if it is wrong …

1 “The dormitory needs a 2 kW heater, so one day costs 48 kWh.”

2 “The campus uses 3,000 kWh a day, so a 30-day month needs 90,000 kWh.”

3 “A bus uses 1.2 kWh per km, so ten buses, each driving one 12 km loop, need 144 kWh.”

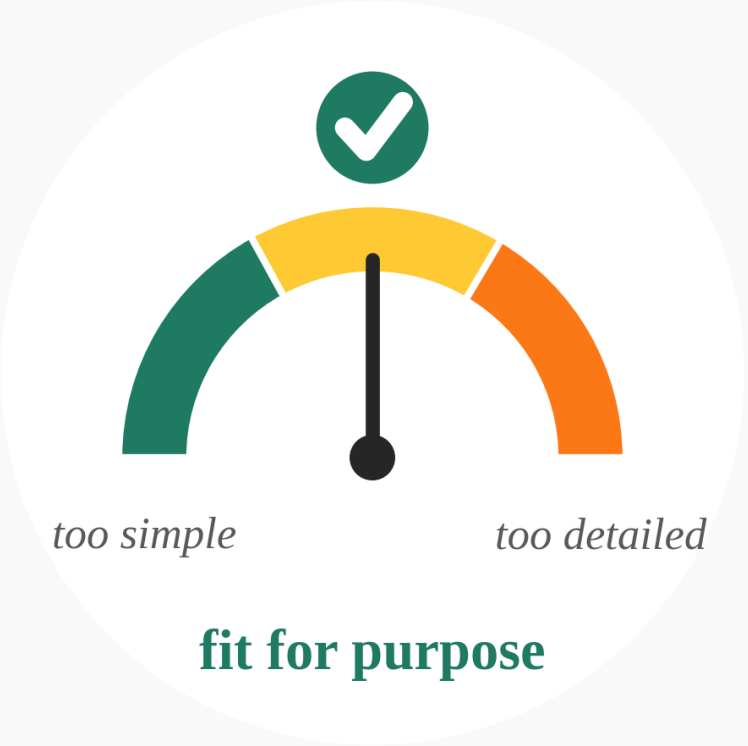
Hint: think about time, weather, and what is the same for every bus.

Spot the Assumptions: Answer Key

Statement	Hidden assumption	One possible rewrite
2 kW heater, 48 kWh per day	Full power for 24 hours, whatever the weather	We assume the heater is always on because we have no thermostat data. If it cycles off, the energy is below 48 kWh.
3,000 kWh a day, 90,000 kWh a month	Every day is an average day	We assume 3,000 kWh every day because we only have an average. If winter days use more, the month exceeds 90,000 kWh.
1.2 kWh/km, ten buses, 144 kWh	Same energy per km for all buses; one loop each	We assume 1.2 kWh/km for all buses because it is the average. If some loops run full, the total exceeds 144 kWh.

Many good rewrites exist. Check that each one names a reason and an effect.

PART 4



revise

Is the Model Good Enough?

Fit for purpose, not for perfection.

More Detail Is Not Always Better

TOO SIMPLE

Misses what matters

Range = 160 ÷ 1.2 for every day of the year.

Winter, hills, and ageing are invisible.

FIT FOR PURPOSE

Answers the question

Separate energy use for summer and winter, plus a safety margin.

Enough to decide: does one charge cover the day?

TOO DETAILED

Cannot be built or checked

Battery chemistry, every traffic light, each driver’s habits.

Needs data we do not have.

Choose the simplest model that answers the question well enough to decide.

Three Sanity Checks

1 **Units.** (u × C) [kWh] ÷ e [kWh/km] gives km, so the range has the right unit.

2 **Order of magnitude.** Is 133 km plausible for a city bus: tens to a few hundred km, not 3 km or 3,000 km?

3 **A known value.** Does it match a datasheet, a test drive, or last month's logs?

Kosky et al. end an energy example with "Is this a reasonable answer?" Ask it every time.

Reasonableness checks: Kosky et al., Ch. 2 (Example 2.4) and Ch. 4 (Example 4.8).

When a Model Misleads

Model	Built for	Used for	What goes wrong
Bus range, e = 1.2 kWh/km	Mild autumn test days	A winter cold snap	Heating raises e; range is overestimated
Solar output profile	A clear summer day	A cloudy winter week	Output is far below the profile
Dormitory load, 2 kW	Mild weather	A week of hard frost	The load is much higher than 2 kW

Every model has a range where its assumptions hold. Know it, and state it.

Build, Check, Revise

1

Frame

Problem statement and variables.

2

Build

A first, simple model with units.

3

Check

Units, size, known values.

4

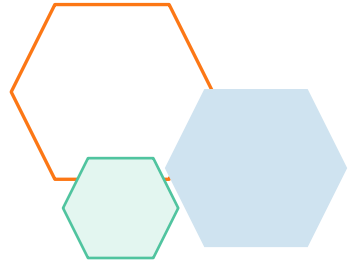
Revise

Fix assumptions; add detail only if needed.

repeat until good enough for the decision

Preview, week 11: sensitivity asks "what if this assumption is wrong, and by how much?"

Key Takeaways



A model simplifies

Built for one purpose: it keeps what matters and drops the rest.

Four types

Physical, graphical, mathematical, and computational.

Four parts

Inputs, parameters, relationships, and outputs, all with units.

Assumptions shape answers

Ignored, fixed, or left outside: write each one down.

Fit for purpose

The simplest model that answers the question well enough to decide.

Check, then revise

Units, order of magnitude, known values; then iterate.

Independent Study: Milestone 2

1. Write your problem framing document: problem statement, performance variables, demands and wishes.
2. Draw your model as a block: inputs, parameters, relationships, outputs, each with units.
3. List at least eight assumptions: what we assume, why, and what changes if it is wrong.
4. Mark each assumption safe, risky, or needs data.

Team deliverable: Milestone 2, the problem framing document and the list of modelling assumptions. Next week we start putting numbers into the model.



Thank You — Questions?

COMING UP NEXT

Lecture 6: Quantitative Reasoning I

Demand and capacity, rates and flows, and order-of-magnitude estimation.